N+ - Network Fundamentals and building network

❖ Network Topologies

[Q-1] Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

[ANS]

- draw topology start , bus , mash

Topology	Example Application/Service	Reason
Star	WhatsApp	Central server manages communication between users.
Bus	Printer-sharing network	All devices share a single communication channel.
Mesh	IPL streaming network	Multiple paths provide reliability and fault tolerance.

[Q-2] Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

[ANS]

Topology	Cost	Reliability	Scalability	Best For
Star	Medium	High	High	Offices, social media apps
Bus	Low	Low	Low	Small networks
Ring	Medium	Medium	Medium	Controlled enterprise networks
Mash	High	Very High	High	Streaming, cloud services, internet backbone

[Q-3] Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

[ANS]

- A client–server (centralized with regional servers)** topology is most suitable for a multiplayer game like PUBG or Free Fire.

  - It improves speed by connecting players to the nearest regional server, reducing lag.
  - It increases reliability because servers handle game data consistently and prevent cheating.
  - Cost is moderate, as multiple servers are needed, but it is scalable and easier to manage.

**[Q-4] Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

Hint: Consider factors like cost, ease of troubleshooting, and future expansion.**

[ANS]

- Recommended topology: Star topology

  - Reason (2–3 lines):
    - Star topology is best because all 8 computers connect to a central switch, making it easy to manage and troubleshoot. If one computer fails, the rest still work. It is also easy to expand by adding more devices.

- Simple diagram:

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            PC1
             |
PC2 — Switch — PC3
 |           |           |
PC4       PC5       PC6
             |
            PC7
             |
            PC8
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